Ch. 7 Cram Sheet: Water Cycle and Watersheds

Exam Fri Nov 6, 40 questions. Know: terms, cycle order, watershed, freshwater numbers.

FRESHWATER NUMBERS

- Salt water (oceans): **~97%**. Fresh water: **~2.5%** (not "most").
- Of the fresh: **~69%** glaciers/ice caps, **~30%** groundwater, **<1%** surface water.

CYCLE ORDER

No starting point; many paths.

1. **Evaporation:** liquid to vapor, sun's heat, any temperature.
2. **Transpiration:** plants release vapor from leaves.
3. **Condensation:** vapor cools to droplets; makes clouds.
4. **Precipitation:** rain, snow, sleet, hail.
5. **Runoff:** flows over land to streams and rivers.
6. **Infiltration:** soaks into soil; can recharge groundwater.

RESIDENCE TIME

How long a molecule stays in one place.

Atmosphere	~9 days
Rivers	~2 weeks
Ocean	~3,200 years
Ice	thousands of years

GROUNDWATER

- **Aquifer:** rock or sediment layer that holds and moves groundwater.
- **Water table:** top of the fully saturated zone. Rises and falls. **NOT flat.**
- **Recharge:** water added to an aquifer by infiltration.

WATERSHEDS

- Land area draining to one outlet (river, lake, sea).
- Edge = **divide** (ridge or high ground).
- Fertilizer, pollution, paving upstream affect the outlet.
- Paving: less infiltration, more runoff, streams rise faster after storms.

Runoff = Precipitation – Infiltration – Evaporation

(Ignores plants and storage.)

MS. REYES'S WARNINGS

- Transpiration = plants. Evaporation = surfaces.
- Water table is not flat.
- The cycle does not start at the ocean.
- Fresh water is ~2.5%, not "most."